

P2FLOW: PHONEME-AWARE PROGRESSIVE FLOW MATCHING FOR EXTREME SPEECH SUPER-RESOLUTION

Ningyuan Yang^{1*}, Yize Li^{2*}, Pu Zhao², Diego A. Cuji¹, Kanad Sarkar³,
Ryan M. Corey⁴, Xue Lin², Andrew C. Singer¹

¹Stony Brook University, NY, US ²Northeastern University, MA, US ³UIUC, IL, US ⁴UIC, IL, US

ABSTRACT

Generative models have recently demonstrated considerable promise in speech super-resolution (SSR). Nevertheless, the majority of existing work has concentrated on standard or versatile SSR configurations, leaving the extreme setting with severely limited spectral inputs largely unexplored. In this regime, current approaches exhibit marked performance degradation, underscoring the need for dedicated solutions. To bridge this gap, we introduce P2Flow, a phoneme-aware progressive flow matching (FM) framework designed for extreme SSR with three main strategies. First, our model leverages phonetic information to reconstruct missing spectral components. Furthermore, it employs a progressive architectural design that hierarchically restores distinct frequency regions. Finally, we incorporate post-training of the vocoder to enhance overall waveform fidelity. Extensive experiments are conducted on the TIMIT and VCTK datasets under both 1 kHz to 16 kHz and 2 kHz to 16 kHz settings, demonstrating that P2Flow yields state-of-the-art results across multiple evaluation metrics.

Index Terms— Flow Matching, Generative Models, Phoneme Awareness, Speech Super-Resolution.

1. INTRODUCTION

Speech super-resolution (SSR) [1, 2], historically termed as speech bandwidth extension [3], aims to reconstruct high-resolution (HR) speech from a low-resolution (LR) input. Early approaches [4, 5, 6, 7, 8, 9] have employed discriminative models to learn direct LR-to-HR mappings, yet often yielded over-smoothed spectra with limited high-frequency (HF) details. More recent generative paradigms [10, 11, 12] have demonstrated superior HF restoration: generative adversarial networks (GANs) [13, 14, 15, 16, 17] improve perceptual realism via adversarial training; diffusion models [18, 19, 20, 21, 22] achieve high-fidelity reconstruction through iterative denoising; flow matching (FM) [23, 24, 25, 26, 27] offers efficient sampling via continuous-time dynamics; and Schrödinger bridges [28, 29, 30] learn stochastic transport between LR and HR distributions.

However, existing research has predominantly focused on standard SSR settings, with moderately reduced input sampling rates (e.g., input→output: 8 kHz→16 kHz), or versatile settings (e.g., arbitrary input rates→48 kHz) [19, 21, 24]. In contrast, the problem of extreme SSR (e.g., 1 or 2 kHz→16 kHz), wherein only severely limited spectral information is available, has received considerably less attention, despite its practical significance in applications such as bone-conduction sensing [14, 31]. To systematically investigate the limitations of current approaches, we first evaluate several representative SSR methods on VCTK [32] with a fixed 16 kHz target

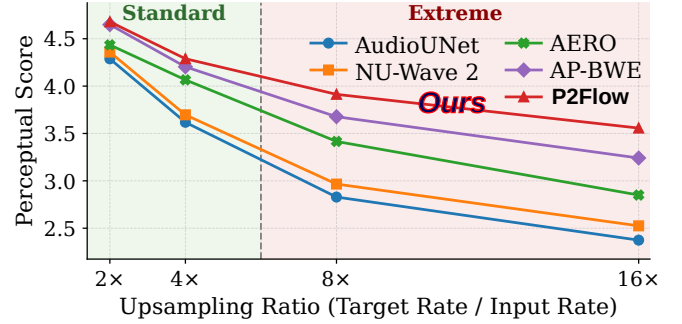


Fig. 1. Perceptual Score for SSR on VCTK with target rate 16 kHz.

rate under varying input rates. As illustrated in Fig. 1, reconstruction performance degrades notably as upsampling ratio increases, underscoring the substantial difficulty of HF restoration when acoustic and phonetic cues are largely insufficient for extreme SSR.

To address these limitations, we propose **P2Flow**, a **Phoneme-aware Progressive Flow** matching framework for extreme SSR. Building upon few-step FM [23], we introduce three key strategies: phoneme guidance, progressive design, and vocoder post-training. (i) Specifically, recognizing that phoneme identity encodes strong cues about harmonic structure and spectral envelope [33, 34], we first integrate a HuBERT-based phoneme classifier into the model. It conditions the FM process on frame-level phoneme posterior distributions to guide HF reconstruction. (ii) In the extreme SSR situation, direct recovery of the largely missing spectral content is particularly difficult. Motivated by image super-resolution [35, 36], we therefore adopt a progressive modeling scheme that decomposes the reconstruction task across frequency bands, enabling HF estimation to leverage lower-frequency representations hierarchically. (iii) To mitigate errors from mel estimation to waveform synthesis, we additionally post-train the vocoder with adversarial learning. Thorough experiments on the TIMIT [37] and VCTK [32] datasets show that P2Flow consistently outperforms representative baselines in spectral fidelity, perceptual speech quality, and intelligibility. Ablation studies further confirm the individual contribution of each proposed component. Overall, P2Flow has demonstrated superiority over other approaches in both standard and extreme settings, as presented in Fig. 1. Our contributions are summarized as follows:

- We identify the overlooked performance degradation of existing SSR methods under extreme band-limited settings.
- We introduce P2Flow, a phoneme-aware progressive FM framework for extreme SSR, comprising phoneme guidance, progressive frequency modeling, and vocoder post-training.
- Comprehensive experiments on TIMIT and VCTK demonstrate state-of-the-art performance and validate each strategy.

*Equal contribution. Code is available here.

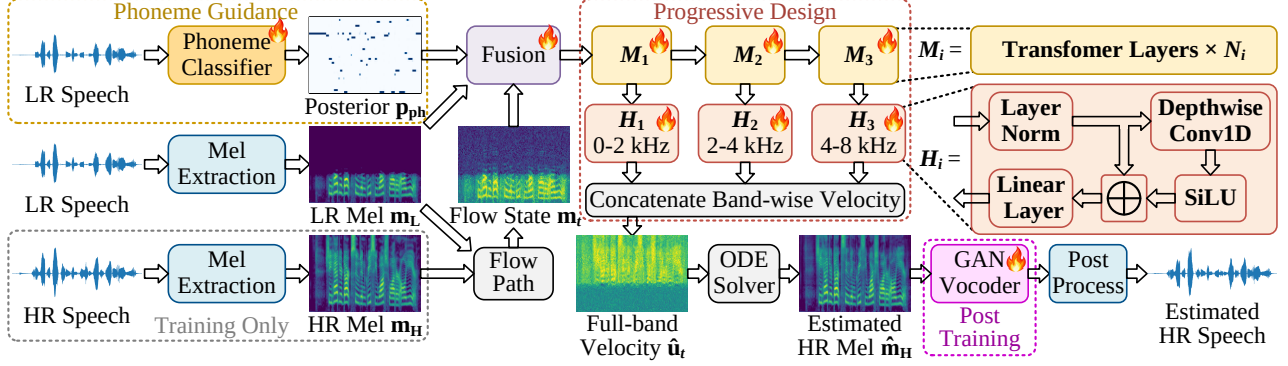


Fig. 2. Overview of the proposed P2Flow framework for HR speech restoration from LR speech. Training follows three steps: (1) phoneme classifier training, (2) FM training with the other modules frozen, and (3) vocoder post-training using estimated mel-spectrograms.

2. FLOW MATCHING PRELIMINARIES

Let $\mathbf{x} \in \mathbb{R}^d$ denote a generic data representation. FM [11, 23] defines a probability path p_t , $t \in [0, 1]$, that transports a source distribution p_0 to a target distribution p_1 . A flow state $\mathbf{x}_t \sim p_t$ evolves according to a time-dependent velocity field v_θ as $\frac{d\mathbf{x}_t}{dt} = v_\theta(\mathbf{x}_t, t)$. FM learns this velocity field such that the induced trajectories follow the prescribed probability path. For a Gaussian path with mean μ_t and standard deviation σ_t , the flow state is sampled as $\mathbf{x}_t = \mu_t + \sigma_t \epsilon$, where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Differentiating with respect to t gives the target velocity $\mathbf{u}_t = \dot{\mu}_t + \dot{\sigma}_t \epsilon$. The predicted velocity field $v_\theta(\mathbf{x}_t, t)$ is trained to match this target by

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, \epsilon} [\|\mathbf{u}_t - v_\theta(\mathbf{x}_t, t)\|_2^2]. \quad (1)$$

At inference, a sample $\mathbf{x}_0 \sim p_0$ is transported toward the target distribution by integrating the learned ordinary differential equation (ODE) $\frac{d\mathbf{x}_t}{dt} = v_\theta(\mathbf{x}_t, t)$ from $t = 0$ to $t = 1$, yielding $\mathbf{x}_1 \sim p_1$.

3. PROPOSED METHOD

P2Flow consists of three complementary components for extreme SSR: (i) a phoneme-aware classifier for lingual conditioning; (ii) progressive hierarchical design in FM to simplify wide-band reconstruction, and (iii) vocoder post-training to reduce mel-to-waveform reconstruction errors. The system pipeline is illustrated in Fig. 2.

3.1. Phoneme Classifier

To provide linguistic guidance for HF restoration [34], we first train a phoneme classifier consisting of a HuBERT encoder [38] and a linear head for frame-level phoneme prediction. Following the standard TIMIT 61-to-39 phone mapping [39], we merge acoustically similar variants into 39 classes. The classifier is trained on speech degraded identically to the SSR input. Since the HuBERT encoder expects 16 kHz waveforms, the LR signal $\mathbf{s}_L$ is first resampled to 16 kHz and normalized, yielding $\tilde{\mathbf{s}}_L$ before feature extraction. We temporally interpolate the HuBERT features from a 20 ms stride to match the 16 ms mel-spectrogram frame stride, then apply the linear head to obtain frame-level phoneme posterior probabilities by

$$\mathbf{p}_{\text{ph}} = \text{softmax}(C_{\text{ph}}(\mathcal{I}(\text{Enc}(\tilde{\mathbf{s}}_L)))) \in \mathbb{R}^{B \times 39 \times T}, \quad (2)$$

where $\text{Enc}(\cdot)$ denotes the HuBERT encoder, $\mathcal{I}(\cdot)$ represents temporal interpolation, $C_{\text{ph}}(\cdot)$ is the linear classification head, B is the batch size, and T is the number of mel-spectrogram frames. The classifier is then frozen during SSR training, while $\mathbf{p}_{\text{ph}}$ provides frame-level linguistic conditioning to the progressive FM model.

3.2. Progressive Modeling

Extreme SSR requires reconstructing a largely missing frequency range, which is particularly challenging. We therefore decompose this difficult mapping into progressive frequency-region modeling, allowing later modules to build on increasingly refined representations. Let $\mathbf{m}_L, \mathbf{m}_H \in \mathbb{R}^{B \times F \times T}$ denote the LR and HR mel-spectrograms, where F is the number of mel-frequency bins. We define $\mu_t = t\mathbf{m}_H + (1-t)\mathbf{m}_L$ and $\sigma_t = 1 - (1 - \sigma_{\min})t$, and sample the flow state as $\mathbf{m}_t = \mu_t + \sigma_t \epsilon$, where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The corresponding target velocity is $\mathbf{u}_t = (\mathbf{m}_H - \mathbf{m}_L) - (1 - \sigma_{\min})\epsilon$. The flow state, LR condition, and phoneme posterior are then concatenated and linearly projected as $\mathbf{h}_0 = W_{\text{in}}[\mathbf{m}_t; \mathbf{m}_L; \mathbf{p}_{\text{ph}}]$. As shown in Fig. 2, our design is based on a few-step FM [23] with progressive phoneme awareness in the mel-spectrogram domain.

Three sequential modules hierarchically refine the embedded representation as $\mathbf{h}_k = M_k(\mathbf{h}_{k-1})$, $k \in \{1, 2, 3\}$. We use the 4/2/2-layer Transformer for M_1 , M_2 , and M_3 , respectively, with a hidden dimension of 1024. Each layer in the Transformer block consists of time-conditioned adaptive RMS normalization, followed by a 16-head self-attention (64-dimensional heads) incorporating query-key RMS normalization and rotary positional embeddings, and then a GEGLU feed-forward network with residual connections. The reversed pyramid design allocates more capacity to M_1 since the first module initially infers spectral structure from the most limited information, while subsequent modules benefit from progressively abundant features. Each M_k is followed by a lightweight velocity head H_k that predicts the target velocity in the lower-frequency (0–2 kHz), middle-frequency (2–4 kHz), or upper-frequency (4–8 kHz) mel region as $\hat{\mathbf{u}}_t^{(k)} = H_k(\mathbf{h}_k)$. Specifically, each head first applies layer normalization, then a depthwise temporal convolution followed by a SiLU activation with a residual connection, and finally a linear projection, as shown in Fig. 2. The three predictions form the full 80-bin velocity field, optimized via

$$\mathcal{L}_{\text{FM}} = \sum_{k=1}^3 \left\| \hat{\mathbf{u}}_t^{(k)} - \mathbf{u}_t^{(k)} \right\|_2^2. \quad (3)$$

During inference, we initialize $\mathbf{m}_0 = \mathbf{m}_L + \sigma \epsilon$ and integrate $v_\theta(\mathbf{m}_t, t, \mathbf{m}_L, \mathbf{p}_{\text{ph}})$ from $t = 0$ to 1 to estimate the HR mel-spectrogram $\hat{\mathbf{m}}_H$.

3.3. Vocoder Post-training

For 16 kHz extreme SSR, we adopt the SpeechBrain HiFi-GAN vocoder [40] for mel-to-waveform synthesis. Since it is pretrained on natural mel-spectrograms, a distribution mismatch arises when

Table 1. 16 kHz extreme SSR performance on TIMIT. Lower ($\downarrow$) is better for LSD metrics; higher ($\uparrow$) is desirable for ViSQOL and STOI.

Model	1 kHz $\rightarrow$ 16 kHz					2 kHz $\rightarrow$ 16 kHz				
	LSD $\downarrow$	LSD-LF $\downarrow$	LSD-HF $\downarrow$	ViSQOL $\uparrow$	STOI $\uparrow$	LSD $\downarrow$	LSD-LF $\downarrow$	LSD-HF $\downarrow$	ViSQOL $\uparrow$	STOI $\uparrow$
Input	4.550	0.723	4.694	1.498	0.663	4.234	0.195	4.525	2.355	0.791
AudioUNet [4]	2.549	0.561	2.626	2.440	0.752	2.436	0.242	2.600	3.036	0.834
NU-Wave 2 [19]	2.129	0.482	2.193	2.458	0.695	1.690	0.354	1.799	3.049	0.797
UDM+ [20]	1.656	0.355	1.707	2.174	0.629	1.456	0.254	1.552	2.910	0.782
AERO [13]	1.440	0.732	1.476	2.778	0.755	1.285	0.557	1.359	3.268	0.847
EBEN [14]	1.126	0.569	1.152	3.139	0.796	1.057	0.584	1.105	3.603	0.871
AP-BWE [15]	1.069	0.343	1.110	3.155	0.800	0.998	0.274	1.060	3.647	0.875
FLowHigh [23]	1.105	0.724	1.124	3.149	0.799	1.017	0.195	1.084	3.712	0.873
UniverSR [24]	1.216	0.365	1.251	2.841	0.761	1.114	0.263	1.186	3.325	0.853
P2Flow (ours)	1.026	0.723	1.041	3.434	0.849	0.935	0.195	0.997	3.954	0.908

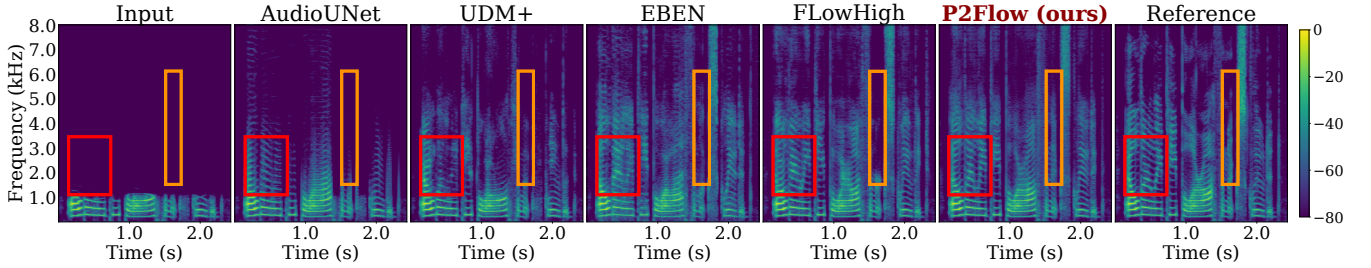


Fig. 3. Spectrogram comparison on TIMIT under the 2 kHz $\rightarrow$ 16 kHz SSR setting. Colored boxes highlight noticeable spectral differences.

decoding mel-spectrogram estimates from the FM model. We therefore post-train the generator on fixed mel estimates while freezing the FM model. A discriminator is jointly trained, while the generator is optimized using standard GAN-vocoder objectives [40, 41, 42, 43], including adversarial, feature-matching, and multi-resolution short-time Fourier transform (MR-STFT) losses, as

$$\mathcal{L}_G = \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{feat}} \mathcal{L}_{\text{feat}} + \lambda_{\text{STFT}} \mathcal{L}_{\text{MR-STFT}}. \quad (4)$$

Specifically, the adversarial and feature-matching losses are given as

$$\mathcal{L}_{\text{adv}} = \sum_k \|D_k(\hat{\mathbf{y}}) - 1\|_2^2. \quad (5)$$

$$\mathcal{L}_{\text{feat}} = \frac{1}{N} \sum_{k,l} \|D_k^{(l)}(\hat{\mathbf{y}}) - D_k^{(l)}(\mathbf{y})\|_1. \quad (6)$$

where $\hat{\mathbf{y}}$ and $\mathbf{y}$ are the generated and target waveforms, D_k is the k -th sub-discriminator, $D_k^{(l)}$ its l -th intermediate feature map, and N the total number of feature maps. The MR-STFT loss is

$$\mathcal{L}_{\text{MR-STFT}} = \frac{1}{R} \sum_{r=1}^R \left(\mathcal{L}_{\text{SC}}^{(r)} + \mathcal{L}_{\text{MAG}}^{(r)} \right), \quad (7)$$

where $\mathcal{L}_{\text{SC}}^{(r)}$ and $\mathcal{L}_{\text{MAG}}^{(r)}$ denote the spectral-convergence and log-magnitude losses at resolution r , respectively. They are defined as

$$\mathcal{L}_{\text{SC}}^{(r)} = \frac{\|S_r(\mathbf{y}) - S_r(\hat{\mathbf{y}})\|_F}{\|S_r(\mathbf{y})\|_F}, \quad (8)$$

$$\mathcal{L}_{\text{MAG}}^{(r)} = \|\log S_r(\mathbf{y}) - \log S_r(\hat{\mathbf{y}})\|_1, \quad (9)$$

where R is the number of STFT resolutions, $S_r(\cdot)$ denotes the STFT magnitude at resolution r , and $\|\cdot\|_F$ is the Frobenius norm. The adversarial and feature-matching losses improve waveform realism and training stability, while the MR-STFT loss preserves spectral structure across multiple resolutions, together reducing errors from mel estimation to vocoder waveform synthesis.

4. EXPERIMENTS

4.1. Experimental Setup

4.1.1. Training

All models are trained and evaluated under identical 1 kHz $\rightarrow$ 16 kHz and 2 kHz $\rightarrow$ 16 kHz SSR settings on TIMIT [37] using the official train/test split and VCTK [32] using a 100/8-speaker train/test split. VCTK phoneme labels are generated with Montreal Forced Aligner (MFA) [44] and mapped to the TIMIT 39-phone set. For each mel frame, we take its temporal midpoint and assign the phoneme label whose aligned time interval contains that point. LR inputs are generated by low-pass filtering the 16 kHz ground truth with a Chebyshev Type-I filter and then downsampling to 1 or 2 kHz. We utilize 80-bin mel-spectrograms with a 1024-sample frame length and 256-sample hop length. The phoneme classifier is trained for 20 epochs, while the FM model is trained for 400k iterations with a batch size of 128 and a learning rate of 3×10^{-4} . The vocoder is fine-tuned for 20 epochs with $(\lambda_{\text{adv}}, \lambda_{\text{feat}}, \lambda_{\text{STFT}}) = (1, 10, 1)$ using three MR-STFT resolutions specified by (FFT size, hop size, window size): (512, 128, 512), (1024, 256, 1024), and (2048, 512, 2048). The FM model is evaluated using 5 sampling steps as the default. We follow the same post-processing strategy in FLowHigh [23] to preserve the reliable low-frequency (LF) spectrum. All experiments are conducted on a single NVIDIA RTX A6000 GPU.

4.1.2. Evaluation

Overall spectral distortion is quantified using log-spectral distance (LSD) [21] (lower is better). We further report LSD-LF and LSD-HF over the preserved LF and missing HF regions. We adopt the virtual speech quality objective listener (ViSQOL) [45] and short-time objective intelligibility (STOI) [46] to evaluate perceptual speech quality and intelligibility, respectively, where higher scores are better.

Table 2. 16 kHz extreme SSR performance on VCTK. Lower ($\downarrow$) is better for LSD metrics; higher ($\uparrow$) is desirable for ViSQOL and STOI.

Model	1 kHz $\rightarrow$ 16 kHz					2 kHz $\rightarrow$ 16 kHz				
	LSD $\downarrow$	LSD-LF $\downarrow$	LSD-HF $\downarrow$	ViSQOL $\uparrow$	STOI $\uparrow$	LSD $\downarrow$	LSD-LF $\downarrow$	LSD-HF $\downarrow$	ViSQOL $\uparrow$	STOI $\uparrow$
Input	4.002	0.273	4.132	1.779	0.685	3.778	0.185	4.037	2.296	0.808
AudioUNet [4]	2.447	0.228	2.525	2.374	0.774	2.323	0.197	2.482	2.829	0.831
NU-Wave 2 [19]	1.917	0.286	1.977	2.525	0.725	1.681	0.283	1.793	2.966	0.814
AP-BWE [15]	1.172	0.251	1.208	3.241	0.805	1.065	0.231	1.135	3.677	0.876
FLowHigh [23]	1.144	0.273	1.179	3.391	0.826	1.054	0.185	1.124	3.752	0.884
P2Flow (ours)	1.056	0.274	1.088	3.557	0.853	0.979	0.185	1.043	3.913	0.906

4.1.3. Baseline

We compare P2Flow with a broad set of representative SSR methods, including AudioUNet [4] as a discriminative model; AERO [13], EBEN [14], and AP-BWE [15] as GAN-based approaches; NU-Wave 2 [19] and UDM+ [20] as diffusion-based baselines; and FLOWHigh [23] and UniverSR [24] as FM-based methods. For all baselines, we follow the official implementations.

4.2. Main Results

Table 1 compares P2Flow with diverse baselines on TIMIT under two extreme settings. P2Flow achieves the best overall performance across both settings, with the lowest LSD/LSD-HF and the highest ViSQOL and STOI. The pronounced gains demonstrate P2Flow’s effectiveness in recovering severely missing HF spectral content while improving perceptual speech quality and intelligibility. Furthermore, Fig. 3 provides a spectrogram comparison on TIMIT. Compared with FLOWHigh [23], P2Flow more closely matches the ground-truth spectral structure: the red boxes show more accurate spectral-gap reconstruction, while the orange boxes highlight HF components recovered by P2Flow but largely missed by FLOWHigh. Similar improvements are observed on VCTK in Table 2, demonstrating consistent performance gains across datasets. Additionally, P2Flow even maintains its superior performance under the 4 kHz $\rightarrow$ 16 kHz and 8 kHz $\rightarrow$ 16 kHz settings, as shown in Fig. 1. More importantly, its advantage persists across various sampling conditions, confirming its general effectiveness for both standard and extreme SSR.

Table 3. Ablation study of P2Flow on TIMIT (2 kHz $\rightarrow$ 16 kHz).

Progressive Phoneme Post-train			LSD $\downarrow$	ViSQOL $\uparrow$	STOI $\uparrow$
$\times$	$\times$	$\times$	0.989	3.831	0.888
$\checkmark$	$\times$	$\times$	0.975	3.848	0.892
$\checkmark$	$\checkmark$	$\times$	0.962	3.856	0.893
$\checkmark$	$\checkmark$	$\times$	0.950	3.943	0.907
$\checkmark$	$\checkmark$	$\checkmark$	0.935	3.954	0.908

4.3. Ablation Study

To further verify the effectiveness of each proposed component, we conduct an ablation study on TIMIT under the 2 kHz $\rightarrow$ 16 kHz setting. In Table 3, progressive modeling consistently improves all metrics, while phoneme conditioning yields the most distinct gains in perceptual quality and intelligibility. Additionally, vocoder post-training reduces spectral distortion and improves overall quality. The optimal performance is achieved by combining all proposed designs.

Moreover, we investigate the layer allocation across the three Transformer modules M_k , $k \in \{1, 2, 3\}$. As shown in Table 4, the 4/2/2 design achieves the best performance, suggesting that devoting more layers to the earlier module effectively mitigates the greater ambiguity inherent in lower-frequency spectral reconstruction.

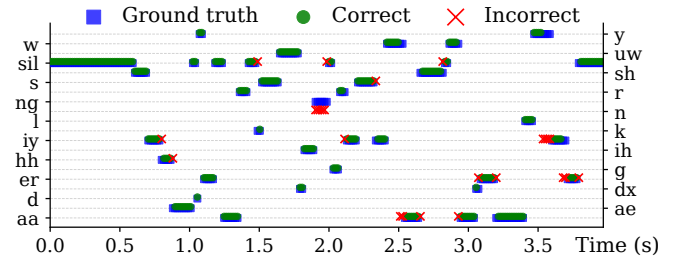
In addition, we report ViSQOL via different sampling steps, achieving scores of 3.946 (1-step), 3.954 (5-step), and 3.954 (10-step). It demonstrates that P2Flow remains highly consistent across inference step counts, thereby underscoring the favorable trade-off between quality and computational cost.

Table 4. Comparison of Transformer layer allocation across the M_1 , M_2 , and M_3 modules in P2Flow on TIMIT (2 kHz $\rightarrow$ 16 kHz).

M_1	M_2	M_3	LSD $\downarrow$	ViSQOL $\uparrow$	STOI $\uparrow$
2-layer	2-layer	4-layer	0.948	3.925	0.906
2-layer	4-layer	2-layer	0.941	3.947	0.907
4-layer	2-layer	2-layer	0.935	3.954	0.908

4.4. Phoneme Analysis

We examine the reliability of phoneme guidance under severely band-limited inputs. The phoneme classifier achieves 82.55% and 75.64% test accuracy for 2 kHz and 1 kHz inputs on TIMIT, respectively. It indicates that substantial phonetic information remains recoverable even under extreme bandwidth reduction. As shown in Fig. 4, most errors occur between acoustically similar phones, such as $\{ng, n\}$ and $\{iy, y\}$, or near phoneme boundaries where transitional cues make classification ambiguous. Rather than relying on hard phoneme labels, our P2Flow instead uses posterior probabilities to retain uncertainty among plausible phones, providing robust phonetic guidance for reconstructing missing HF content.

**Fig. 4.** An example of temporal phoneme predictions on TIMIT.

5. CONCLUSION

This work investigates the underexplored problem of extreme SSR and proposes P2Flow, a phoneme-aware progressive FM framework. P2Flow combines phoneme guidance, progressive spectral generation, and vocoder post-training to improve reconstruction under severe bandwidth constraints. Evaluated on TIMIT and VCTK, P2Flow surpasses competitive baselines by a clear margin in spectral, perceptual, and intelligibility metrics. Moreover, our ablation and phoneme-based investigations confirm the individual contribution and necessity of each proposed component. Future work will investigate the extension of P2Flow to more diverse and realistic bandwidth-degradation scenarios.

6. REFERENCES

- [1] Ningyuan Yang, Yize Li, et al., “A survey of advancing audio super-resolution and bandwidth extension from discriminative to generative models,” *arXiv preprint arXiv:2605.16681*, 2026.
- [2] Sefik Emre Eskimez et al., “Speech super resolution generative adversarial network,” in *IEEE ICASSP*, 2019.
- [3] Kehuang Li et al., “A deep neural network approach to speech bandwidth expansion,” in *IEEE ICASSP*, 2015.
- [4] Volodymyr Kuleshov, S Zayd Enam, and Stefano Ermon, “Audio super resolution using neural networks,” *arXiv preprint arXiv:1708.00853*, 2017.
- [5] Teck Yian Lim, Raymond A. Yeh, et al., “Time-frequency networks for audio super-resolution,” in *IEEE ICASSP*, 2018.
- [6] Sawyer Birnbaum, Volodymyr Kuleshov, et al., “Temporal film: Capturing long-range sequence dependencies with feature-wise modulations,” in *NeurIPS*, 2019.
- [7] Nathanaël Carraz Rakotonirina, “Self-attention for audio super-resolution,” in *IEEE MLSP*, 2021.
- [8] Heming Wang and DeLiang Wang, “Towards robust speech super-resolution,” *IEEE/ACM TASLP*, 2021.
- [9] Viet-Anh Nguyen et al., “Tunet: A block-online bandwidth extension model based on transformers and self-supervised pre-training,” in *IEEE ICASSP*, 2022.
- [10] Jonathan Ho, Ajay Jain, and Pieter Abbeel, “Denoising diffusion probabilistic models,” in *NeurIPS*, 2020.
- [11] Yaron Lipman, Ricky T. Q. Chen, et al., “Flow matching for generative modeling,” in *ICLR*, 2023.
- [12] Linqi Zhou, Aaron Lou, Samar Khanna, and Stefano Ermon, “Denoising diffusion bridge models,” in *ICLR*, 2024.
- [13] Moshe Mandel et al., “Aero: Audio super resolution in the spectral domain,” in *IEEE ICASSP*, 2023.
- [14] Julien Hauret et al., “Eben: Extreme bandwidth extension network applied to speech signals captured with noise-resilient body-conduction microphones,” in *IEEE ICASSP*, 2023.
- [15] Ye-Xin Lu et al., “Towards high-quality and efficient speech bandwidth extension with parallel amplitude and phase prediction,” *IEEE/ACM TASLP*, 2024.
- [16] Yongjoon Lee and Chanwoo Kim, “Wave-u-mamba: an end-to-end framework for high-quality and efficient speech super resolution,” in *IEEE ICASSP*, 2025.
- [17] Ting-Wei Zhang and Shanq-Jang Ruan, “Vm-asr: A lightweight dual-stream u-net model for efficient audio super-resolution,” *IEEE/ACM TASLP*, 2025.
- [18] Junhyeok Lee and Seungu Han, “Nu-wave: A diffusion probabilistic model for neural audio upsampling,” in *Proc. Interspeech*, 2021.
- [19] Seungu Han and Junhyeok Lee, “Nu-wave 2: A general neural audio upsampling model for various sampling rates,” in *Proc. Interspeech*, 2022.
- [20] Chin-Yun Yu, Sung-Lin Yeh, György Fazekas, et al., “Conditioning and sampling in variational diffusion models for speech super-resolution,” in *IEEE ICASSP*, 2023.
- [21] Haohe Liu, Ke Chen, et al., “Audiosr: Versatile audio super-resolution at scale,” in *IEEE ICASSP*, 2024.
- [22] Yuan Fang et al., “Vector quantized diffusion model based speech bandwidth extension,” in *IEEE ICASSP*, 2025.
- [23] Jun-Hak Yun et al., “Flowhigh: Towards efficient and high-quality audio super-resolution with single-step flow matching,” in *IEEE ICASSP*, 2025.
- [24] Woongjib Choi, Sangmin Lee, et al., “Universr: Unified and versatile audio super-resolution via vocoder-free flow matching,” in *IEEE ICASSP*, 2026.
- [25] Jaekwon Im et al., “Saga-sr: Semantically and acoustically guided audio super-resolution,” in *IEEE ICASSP*, 2026.
- [26] Tsun-An Hsieh et al., “Towards real-time generative speech restoration with flow-matching,” in *IEEE ICASSP*, 2026.
- [27] Bowen Zhang et al., “Codecflow: Efficient bandwidth extension via conditional flow matching in neural codec latent space,” *arXiv preprint arXiv:2603.02022*, 2026.
- [28] Zhifeng Kong et al., “A2sb: Audio-to-audio schrodinger bridges,” *arXiv preprint arXiv:2501.11311*, 2025.
- [29] Chang Li, Zehua Chen, et al., “Bridge-sr: Schrödinger bridge for efficient sr,” in *IEEE ICASSP*, 2025.
- [30] Chang Li, Zehua Chen, et al., “Audio super-resolution with latent bridge models,” in *NeurIPS*, 2026.
- [31] Changtao Li et al., “A two-stage approach to quality restoration of bone-conducted speech,” *IEEE/ACM TASLP*, 2023.
- [32] Junichi Yamagishi, Christophe Veaux, and Kirsten MacDonalld, “Cstr vctk corpus: English multi-speaker corpus for cstr voice cloning toolkit (version 0.92),” 2019.
- [33] Nana Hou, Chenglin Xu, Van Tung Pham, et al., “Speaker and phoneme-aware speech bandwidth extension with residual dual-path network,” in *Proc. Interspeech*, 2020.
- [34] Maxime Poli, Emmanuel Chemla, et al., “Improving spoken language modeling with phoneme classification: A simple fine-tuning approach,” in *EMNLP*, 2024.
- [35] Yifan Wang et al., “A fully progressive approach to single-image super-resolution,” in *CVPRW*, 2018.
- [36] Syed Waqas Zamir et al., “Multi-stage progressive image restoration,” in *CVPR*, 2021.
- [37] John S Garofolo, Lori F Lamel, et al., “Getting started with the darpa timit cd-rom: An acoustic phonetic continuous speech database,” *NIST, Gaithersburg, MD*, 1988.
- [38] Wei-Ning Hsu, Benjamin Bolte, et al., “Hubert: Self-supervised speech representation learning by masked prediction of hidden units,” *IEEE/ACM TASLP*, 2021.
- [39] K-F Lee et al., “Speaker-independent phone recognition using hidden markov models,” *IEEE/ACM TASLP*, 1989.
- [40] Jungil Kong, Jaehyeon Kim, and Jaekyoung Bae, “Hifi-gan: Generative adversarial networks for efficient and high fidelity speech synthesis,” in *NeurIPS*, 2020.
- [41] Sang-gil Lee, Wei Ping, et al., “Bigvgan: A universal neural vocoder with large-scale training,” in *ICLR*, 2023.
- [42] Ryuichi Yamamoto, Eunwoo Song, and Jae-Min Kim, “Parallel wavegan: A fast waveform generation model based on generative adversarial networks with multi-resolution spectrogram,” in *IEEE ICASSP*, 2020.
- [43] Geng Yang et al., “Multi-band melgan: Faster waveform generation for high-quality text-to-speech,” in *IEEE SLT*, 2021.
- [44] Michael McAuliffe et al., “Montreal forced aligner: Trainable text-speech alignment using kaldii,” in *Proc. Interspeech*, 2017.
- [45] Michael Chinen, Felicia SC Lim, et al., “Visqol v3: An open source production ready objective speech and audio metric,” in *IEEE QoMEX*, 2020.
- [46] Cees H Taal, Richard C Hendriks, et al., “A short-time objective intelligibility measure for time-frequency weighted noisy speech,” in *IEEE ICASSP*, 2010.